

PROPAGATION IDENTITY

The Missing Primitive for Physical AI

WHITEPAPER

Technical Edition for Protocol Engineers, AI Safety Researchers, and Regulators

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*“The journey is the proof.”
— GNS Protocol Design Principle*

Abstract

Embodied artificial intelligence has crossed an industrial threshold. Vision-Language-Action (VLA) models now drive humanoid robots in production deployments at Mercedes-Benz, Hyundai, GXO, BMW and Foxconn. Industrial-arm platforms operate under Google’s Intrinsic Flowstate runtime across the FANUC, KUKA and Universal Robots ecosystems. None of these systems carry a cryptographic record of what model decided what physical action, under whose authority, in which jurisdiction. Existing provenance primitives — TLS, code signing, C2PA, SLSA, SPIFFE, TPM, DIDs, AARM — solve neighbouring problems and stop at the boundary where a software decision becomes a physical motion.

This paper introduces propagation identity: the property of an agentic physical system whereby every action in the physical world is cryptographically traceable back through the full chain of decisions that produced it — human intent, model inference, runtime planning, actuator command — with each layer signing its own contribution and inheriting constrained authority from the layer above. Propagation identity is positioned as Layer 5 in the emerging provenance stack, composing rather than displacing the layers below.

We define the primitive formally, present a four-layer signing chain (operator → model → runtime → actuator), bind each action to an H3 hexagonal cell and a sealed temporal epoch, and show how the (cell, epoch, public_key) triple maps natively onto the MobyDB geospatial-native database. We describe the ULissy language construct propagates from that makes the chain a compile-time guarantee rather than a runtime convention. We map the requirement to the Machinery Regulation 2023/1230 (effective January 2027), the EU AI Act high-risk regimes (August 2026 / August 2027), and the corresponding US frameworks. We propose an IETF standardization path as an extension of the GNS Trajectory Identity Protocol (TriP) Internet-Draft.

The paper is intended as a reference for protocol engineers integrating physical-AI systems with regulatory frameworks, for AI safety researchers studying the decision-to-action boundary, and for the certification bodies (TÜV, Bureau Veritas, DEKRA) that will be asked, beginning in 2027, to attest that a robot acted as authorized.

Executive Summary

Seven findings frame the argument of this paper. Each is developed in detail in the indicated section.

1. The decision-to-action boundary is the location of the gap.

Software-agent provenance frameworks (Google AP2, Prova, Anjuna, AARM) and physical-attestation primitives (TPM, Intel TDX, secure boot) are both mature in their respective domains. Neither extends across the boundary where a model output becomes a motor command. (§1, §6)

2. Existing provenance primitives do not propagate.

TLS chains terminate at the endpoint. Code signing terminates at the binary. C2PA terminates at the media file. SLSA and in-toto terminate at the build artifact. SPIFFE/SPIRE terminates at the workload. None of these carries identity into a physical action. (§2)

3. The 2026–2027 regulatory window forces the primitive.

EU AI Act Annex III obligations apply 2 August 2026. The Machinery Regulation 2023/1230, which classifies AI software as a safety component and requires logs of all interactions to be maintained for traceability, becomes enforceable 20 January 2027. EU AI Act Annex I obligations — covering AI embedded in regulated products including machinery — apply 2 August 2027. The Machinery Regulation describes propagation identity in regulatory prose without naming it. (§5)

4. Industry-state evidence: zero of ten major embodied platforms ship cryptographic action provenance.

A landscape survey conducted in April 2026 of Agility Digit, Appteronik Apollo, Boston Dynamics Atlas, Figure 03, 1X NEO, Tesla Optimus, XPENG IRON, Unitree G1/H1, and the Intrinsic Flowstate industrial arm ecosystem (FANUC, KUKA, Universal Robots) finds that none cryptographically signs runtime actions. Only Intuitive Surgical da Vinci operates under an audit-trail regime, and that regime is pre-cryptographic, designed for centralized hospital infrastructure under 21 CFR Part 11. (§4)

5. Gemini Robotics is structurally a multi-layer identity system that does not sign its layers.

Gemini Robotics 1.5 / ER 1.6 is a dual-model architecture: an embodied-reasoning planner produces natural-language reasoning traces interleaved with motor commands, and a vision-language-action policy executes them. Cross-embodiment Motion Transfer means the same skill executes on heterogeneous hardware bodies without retraining. The decision layers are already discrete and interpretable; their absence from cryptographic chains is therefore a deliberate omission, not an architectural impossibility. (§4.2)

6. Propagation identity is the composition layer, not a replacement.

The primitive can be defined and implemented entirely as a composition of existing layers: TPM-rooted hardware attestation at the actuator, SPIFFE-style workload identity at the runtime, AARM-style action provenance at the model, and a delegation certificate at the operator — connected by a four-signature breadcrumb anchored to an H3 cell and a sealed epoch. This positions GNS, GEIANT, MobyDB and ULissy as the layer the existing stack has been building toward. (§7–§14)

7. Google’s Intrinsic, post-merger, is a partner shape, not a competitor shape.

The Intrinsic / Gemini Robotics / Google Cloud stack is optimized for platform adoption, Gemini token consumption, and cloud margin. Compliance and runtime attestation are orthogonal to that revenue model. Intrinsic structurally will not build propagation identity. The natural integration path is GNS-AIP SDK as a drop-in compliance layer for Flowstate deployments in

regulated markets, with go-to-market routed through OEM partners (FANUC, KUKA, Universal Robots) and EU automotive customers (Mercedes-Benz, BMW, Comau). (§17)

PART I

The Problem

1. The Identity Collapse at the AI/Robotics Boundary

1.1 Thesis

A pure software agent can act without cryptographic propagation because its harms are reversible and bounded. A pure pre-programmed robot can act without it because its program is deterministic and its envelope is constrained. The moment an artificial-intelligence model decides what a robot does in an open physical environment, the identity chain becomes unsolvable by every existing primitive — yet regulators, insurers, and courts are about to demand exactly that chain.

1.2 The Two Classical Regimes That No Longer Apply

Two classical regimes have, until recently, governed accountability for systems that act in the physical world.

The deterministic-machine regime applies to industrial automation as it has existed since the 1970s. A welding robot follows a hand-coded path. A warehouse conveyor follows a hand-coded interlock graph. The behaviour of the machine is fully specified by its program; the program is fully specified by its source code; the source code is reviewed and approved by a named human engineer before commissioning. Regulatory frameworks such as ISO 10218 and the previous Machinery Directive 2006/42/EC are written for this regime. The accountability question — who authorized this action — has a single, statically resolvable answer.

The bounded-software regime applies to enterprise software systems. A trading algorithm executes orders within risk limits. A fraud-detection system flags transactions. Errors are reversible because the harm is informational: positions can be unwound, flags can be retracted, audit logs are centralized in vendor-controlled databases. Frameworks such as SOX, MiFID II and HIPAA are written for this regime. The accountability question is answered through transactional logs in trusted infrastructure.

Embodied AI satisfies neither regime. The behaviour of the machine is no longer specified by its program — it is specified by the runtime inference of a model whose weights were learned from data, whose decisions are not fully predictable, and whose deployment may include

cross-embodiment transfer to hardware the model was not originally trained on. The harms are not informational — a 200-kilogram humanoid in a Mercedes-Benz body shop can crush a worker, weld the wrong panel, or damage a chassis. The audit logs, where they exist at all, are not cryptographically bound to the actions they purport to record. The regulator's question — who authorized this action — has no statically resolvable answer.

1.3 What Has Changed Since 2024

Three concurrent shifts have collapsed the gap between 'software agent' and 'physical machine' from a margin of years to a margin of months.

1. VLA models entered production. Gemini Robotics 1.5 and ER 1.6, Helix 02, the in-house Vision-Language-Action stacks at 1X, Figure and Apptronik, and Tesla's Optimus running on Grok, all moved from research demonstrations to commercial deployments during 2025. Apptronik Apollo is operating in Mercedes-Benz body shops in Germany and Hungary. Agility Digit has crossed 100,000 totes moved in production at GXO. Figure 03 has paying logistics customers. The deployment phase has begun.
2. Cross-embodiment generalization shipped. DeepMind's Motion Transfer, demonstrated in late 2025, allows a skill learned on an ALOHA 2 dual-arm platform to execute on an Apptronik Apollo humanoid without retraining. The skill therefore has an identity trajectory across hardware bodies. No existing primitive signs or tracks that trajectory.
3. Industrial automation re-architected as an open platform. Google's February 2026 absorption of Intrinsic, and the announcement of Flowstate as the cross-vendor robotics runtime — described by its makers as the 'Android for robots' — means that industrial arms are no longer monolithic vendor stacks but heterogeneous compositions of model, runtime, skills library and hardware that may have come from four different organizations. The accountability surface multiplied.

1.4 The Question Regulators Will Ask

In the period between 2 August 2026 (EU AI Act Annex III) and 2 August 2027 (EU AI Act Annex I, including machinery-embedded AI under Regulation 2023/1230), the question that supervisory authorities will be authorized to ask of any embodied-AI deployment is operationally identical to the question already asked of human-operated industrial systems: produce the chain of authority and the record of action. Article 12 of the AI Act and Article 22 of the Machinery Regulation make this explicit.

The chain of authority for an embodied-AI deployment is at least four layers deep — operator delegation, model inference, runtime planning, actuator command — and every layer carries an identity that today goes unsigned. The record of action is, on every commercially deployed platform surveyed in §4, a vendor-controlled log file with no cryptographic binding to the physical motion it claims to describe. The regime that answered the regulator's question for the previous fifty years of industrial automation does not answer it for the regime that is now being deployed.

2. Why Existing Primitives Do Not Propagate

2.1 The Propagation Property

An identity primitive *propagates* if the cryptographic binding it establishes survives the transformation of an input into an output. TLS does not propagate: a TLS-authenticated request produces a server response that carries no record of the requesting identity beyond the TLS session. Code signing does not propagate: a signed binary executes producing outputs that carry no record of which signed binary produced them. C2PA does not propagate: a signed image embedded in a document produces a derivative that, absent explicit republishing, loses the chain. Each of these primitives establishes an authentic point — a session, a binary, a media file — and stops there.

Propagation, as the term is used in this paper, is the property by which an identity binding crosses a transformation boundary and emerges on the far side as a verifiable component of the output. A primitive that propagates allows a verifier to inspect an output and reconstruct the chain of inputs that produced it, with a cryptographic guarantee at each link.

No widely-deployed identity primitive in 2026 propagates across the four transformation boundaries that matter for embodied AI: human intent → delegation, delegation → model inference, model inference → runtime plan, runtime plan → actuator command. Each boundary is the locus of a transformation that an existing primitive could in principle bind, but does not.

2.2 Gap Analysis: Eight Existing Primitives, Eight Stopping Points

The table below summarizes the eight most relevant existing identity and provenance primitives, identifies the boundary at which each terminates, and identifies what each leaves unsigned in an embodied-AI deployment. The columns are derived from the specifications of each primitive as of April 2026.

Primitive	Established Layer	Termination Boundary	What Remains Unsigned in an Embodied-AI Action
TLS / mTLS	Network session	Endpoint of the session	Decision content; downstream actions
Code signing	Software supply chain	Binary artifact	Runtime decisions; physical effects
C2PA	Media provenance	The media file itself	Inferences derived from media; actions taken
SLSA / in-toto	Build-time supply chain	Build artifact	Runtime invocation; actuator commands
SPIFFE / SPIRE	Workload identity	The workload process	Physical-world effects; spatial scope

Primitive	Established Layer	Termination Boundary	What Remains Unsigned in an Embodied-AI Action
TPM / TDX / SEV	Hardware attestation	Boot integrity / enclave state	Inference content; spatial-jurisdictional binding
W3C DIDs / VCs	Decentralized identity	Subject identity	Geospatial scope; runtime delegation chain
AARM / AP2 / Prova	Software-agent action	Software action / payment	Physical actuation; spatial-jurisdictional anchor

Three observations follow from the table.

- The eight primitives, taken together, cover every layer of a modern computing stack from hardware boot through to software-agent action. They do not, taken together, cover physical actuation.
- Two primitives — AARM/AP2/Prova at the software-agent layer, and TPM/TDX at the hardware layer — are structurally close to propagation identity but address only the layers above and below the gap, not the gap itself.
- None of the primitives anchors an action to a verifiable spatial coordinate or a verifiable temporal epoch in a way that maps to the jurisdictional structure of the regulations that will be enforced beginning in 2026 and 2027.

2.3 Why the Gap Has Persisted

The gap between software-agent and physical-actuation provenance has persisted for three reasons, each of which has now been substantially eroded by industry developments.

Performance budgets. Robotics control loops historically operated at 1 kHz with strict latency budgets that excluded any signing operation in the hot path. Modern Ed25519 signing on commodity ARM hardware completes in under 100 μ s, and the natural unit of signed action is the skill-level command (10–100 Hz) rather than the joint-level command (1 kHz), eliminating the budget objection.

Closed vendor stacks. Until 2024, industrial robotics was sold as a closed vertical stack — FANUC programmed FANUC robots with FANUC software. There was no inter-vendor boundary that demanded propagating identity across organizations. The Intrinsic Flowstate platform, the Gemini Robotics multi-vendor deployments, and the cross-embodiment generalization of VLA skills have replaced that closed stack with a heterogeneous one.

No regulatory forcing function. Until the AI Act and the Machinery Regulation, no enforcement regime explicitly required cryptographic traceability of decisions to physical actions. Both regulations now require it in regulatory prose, as documented in §5.

The three conditions that protected the gap have all been removed within an 18-month window. The gap is now a deployment-blocking deficiency, not a future research question.

3. Five Concrete Failure Modes

The argument of §1 and §2 is structural. This section grounds it in five concrete failure modes that an unsigned embodied-AI deployment cannot defend against, and that a propagating identity primitive structurally prevents. Each mode is described with a worked example drawn from a deployment configuration that exists in production as of April 2026.

3.1 VLA Hallucination

Configuration. An Apptronik Apollo humanoid at a Mercedes-Benz body shop is given a high-level instruction: ‘pick the chassis bracket from rack B-3 and place it on the welding fixture.’ The instruction is sent to Gemini Robotics ER 1.6, which produces a natural-language reasoning trace and a sequence of motor primitives. The motor primitives are executed by the underlying VLA policy on the humanoid hardware.

Failure. The VLA policy hallucinates. The reasoning trace from ER 1.6 specified ‘grasp bracket from B-3’; the motor sequence executed by the VLA grasps from B-2, where a worker’s tool tray is located. The action damages the tool tray and triggers a safety stop. The post-incident question is: what specifically did the VLA do that the planner did not authorize?

Why existing primitives do not answer. The runtime log records a sequence of joint commands. It does not cryptographically bind those commands to the planner output that was supposed to authorize them. There is no signed artifact stating ‘these motor primitives are the ones ER 1.6 produced for this instruction at this time on this hardware.’ The investigation devolves into log forensics in a vendor-controlled database — exactly the regime the Machinery Regulation is designed to replace.

3.2 Delegation Escape

Configuration. A Foxconn factory deploys an Intrinsic Flowstate workcell with three Universal Robots arms. The factory operator delegates a scope-limited authority to the workcell: ‘assemble units of SKU 1842, restrict to assembly line 4, between 06:00 and 18:00 local time.’ The delegation is recorded as a configuration entry in Flowstate.

Failure. Through a misconfigured skill-library update, one of the three arms accepts a task to perform a disassembly action — within the same SKU but outside the delegated scope. The task is executed; the resulting unit is non-conforming and enters the supply chain. When the issue is discovered three weeks later, the operator must demonstrate to the customer and the certification body that the disassembly action was outside the delegation. The configuration entry is text in a database; it carries no cryptographic constraint that would have refused the out-of-scope command at runtime.

Why existing primitives do not answer. OAuth scopes, RBAC tokens and SPIFFE workload identities all support scope-limited delegation, but none of them is enforced by the actuator.

The decision to refuse an out-of-scope command must be made at the layer that issues motor primitives, and that layer has no cryptographic input to act on.

3.3 Spatial Violation

Configuration. An Agility Digit unit operates inside a GXO warehouse in Atlanta, GA, USA. The unit is authorized to work on aisles 12 through 18, identified by the warehouse management system. The authorization is policy in the WMS; it is not bound to a verifiable spatial coordinate.

Failure. A pallet of unrelated inventory is moved to aisle 19 by a forklift error. Digit, following its task assignment, walks to aisle 19 to retrieve a tote and operates outside the authorized zone. The action is logged as a normal task completion. Three months later the warehouse undergoes an OSHA inspection and is asked to demonstrate which physical zones the autonomous units operated in over the audit period. The WMS log can produce a list of aisles by name; it cannot produce a cryptographically bound spatial trace.

Why existing primitives do not answer. GPS, RTLS and indoor positioning systems produce coordinate streams. None of them produces a signed coordinate stream bound to the actions the unit took. There is no primitive that binds an action to an H3 hexagonal cell, with the cell stored as a primary key, in a way that makes the spatial trace forensically reconstructable years after the deployment.

3.4 Model Substitution

Configuration. A surgical-assist robot operating under FDA 21 CFR Part 11 audit-trail requirements uses a vision-classification model that has been certified by the manufacturer. The certification record is held by the manufacturer; the runtime log records the model version string.

Failure. A routine over-the-air update silently substitutes a newer, uncertified model variant. The version string in the runtime log is updated to reflect the new model. There is no signed record of which specific model weights were active at the time of any specific procedure. When an adverse event occurs three months post-update, the manufacturer cannot prove which model was active during the procedure in question.

Why existing primitives do not answer. Code signing covers the binary that loads the model. C2PA covers media files. SLSA covers the build artifact. None of them produces, for each individual inference, a cryptographic record of the model weights that produced that inference. The Machinery Regulation 2023/1230 explicitly requires re-conformity assessment when an AI behaviour change is introduced; without per-inference model attestation, 'behaviour change' is not a defined event.

3.5 Sensor Spoofing

Configuration. A Boston Dynamics Atlas unit operating at a Hyundai facility uses an onboard LIDAR and stereo camera as inputs to its perception stack. The perception stack feeds the VLA policy. The actions taken are recorded with timestamps.

Failure. An adversary, exploiting a vulnerability in the sensor firmware, injects synthetic point-cloud frames into the LIDAR stream. The unit, perceiving a non-existent obstacle, executes an emergency-stop manoeuvre that damages a \$200,000 robot arm in the adjacent cell. The post-incident investigation must determine whether the perception that triggered the action was authentic. The sensor firmware was not signing its frames.

Why existing primitives do not answer. TPM-backed sensor attestation exists in research but is not deployed in commercial robotics. C2PA-style media provenance was designed for human-consumed media, not for inference-input streams. There is no commercial primitive that produces, for each VLA decision, a signed record of the perception inputs the decision was based on.

3.6 The Common Structure of the Five Failures

The five failures share a single structural property. In each case, an action in the physical world has occurred, and the question of accountability requires reconstructing the chain of decisions that produced the action. In each case, the chain is not cryptographically reconstructable, because the layer that connects software decisions to physical actuation is unsigned.

Propagation identity, as defined in Part III, addresses all five failure modes through the same structural mechanism: each layer in the decision chain produces a signed contribution that is bound to the layer above by inheritance and to the physical world by an H3 cell and a sealed epoch. The mechanism is the same; the failure modes it prevents are diverse because they share an underlying cause.

PART II

The Landscape

4. The Current State of Embodied Artificial Intelligence

4.1 The Deployment Threshold Has Been Crossed

Until the second half of 2024, embodied artificial intelligence was a research category. Vision-Language-Action models were demonstrated in academic publications and in vendor showcases; humanoid platforms accumulated venture funding without meaningful unit deployments; industrial-arm AI was confined to narrow pilot programmes inside automotive and electronics manufacturers. The transition from that research regime to a deployment regime, dated retrospectively, occurs across an eighteen-month interval bounded by the launch of Gemini Robotics 1.5 in mid-2025 and the Q1 2026 mass-production launch of XPENG IRON. The deployments enumerated below are commercial, paid, and operating in physical environments occupied by human workers as of April 2026.

- Apptронik Apollo humanoids are deployed in Mercedes-Benz body shops in Sindelfingen, Germany and Kecskemét, Hungary, performing parts handling and pre-assembly tasks adjacent to human workers.
- Agility Digit units have crossed an aggregate of more than one hundred thousand totes moved in production at GXO Logistics warehouses in the United States. Paid contracts with Toyota and Mercado Libre have been disclosed.
- Boston Dynamics Atlas, post-CES 2026, has begun a production ramp under Hyundai's commitment, with first commercial deployments scheduled for Q3 2026.
- Figure 03 is operating at BMW and at an unnamed logistics customer; consumer home pilots have been announced for the second half of 2026.
- 1X NEO has opened consumer pre-orders for 2026 home delivery, marking the first humanoid sold directly to retail consumers.
- Tesla Optimus, running on the xAI Grok model, is in limited internal use at Tesla manufacturing facilities.
- XPENG IRON entered Q1 2026 mass production under XPENG's in-house Physical AI stack.
- Unitree shipped approximately 5,500 G1 and H1 humanoids during 2025, with a stated 2026 target of ten to twenty thousand units, predominantly to research and integrator customers.
- The Intrinsic Flowstate runtime, post-Google merger in February 2026, has been extended across the FANUC, KUKA and Universal Robots industrial-arm ecosystems, with Foxconn announcing a joint venture in October 2025 for general-purpose intelligent robots in electronics manufacturing.

The aggregate evidence of these deployments establishes that embodied artificial intelligence has, as of the publication of this paper, crossed the threshold from controlled-pilot to

operational-deployment. The question of whether the regulatory and accountability infrastructure should match this state is therefore not a hypothetical concern about a future technology; it is a present concern about systems that are already operating in environments where human safety, regulatory compliance, and product liability are immediately consequential.

4.2 Gemini Robotics: A Dual-Model Architecture That Already Has Identity Layers

The Gemini Robotics 1.5 stack, jointly developed by Google DeepMind and Intrinsic and used in production deployments under partnerships with Apptроник, Boston Dynamics, and the post-merger Intrinsic Flowstate ecosystem, provides the clearest publicly-documented example of an embodied-AI system whose architecture already contains discrete identity layers without signing them.

The architecture comprises two cooperating models. An embodied-reasoning planner — currently Gemini Robotics ER 1.6 — receives high-level instructions and produces a natural-language reasoning trace interleaved with a sequence of motor primitives. The planner is described by its makers as ‘thinking before acting’: its output is interpretable, separable from the motor commands it produces, and recoverable as a discrete artefact. A vision-language-action policy, comprising the lower-level Gemini Robotics 1.5 model and onboard control software, executes the motor primitives on the physical hardware. The two models communicate through an explicit, recoverable interface.

Cross-embodiment Motion Transfer is a technique demonstrated by DeepMind during 2025 in which a skill — such as folding a piece of laundry, or grasping an object from a cluttered surface — is learned on one hardware platform (typically the ALOHA 2 dual-arm research platform) and executed without retraining on a different hardware platform (such as an Apptроник Apollo humanoid). The skill, in this regime, is not bound to the hardware on which it was learned. It has an identity trajectory across hardware bodies. This property is a strong engineering result and an unsigned runtime event.

The implication for accountability is direct. A given physical action — for example, a humanoid grasping a chassis bracket and placing it on a fixture in a Mercedes-Benz body shop — is the composition of at least four discrete decisions: an operator’s instruction (delegation), a planner’s reasoning trace (model inference), a VLA policy’s motor sequence (runtime), and a hardware actuator’s motor command (physical action). Each layer is architecturally discrete. Each layer is recoverable as a separable artefact. None of the four layers, in any production deployment surveyed for this paper, is signed.

4.3 Landscape Snapshot

The following table, assembled from public announcements and first-party technical documentation through April 2026, summarizes the major embodied-AI platforms, their VLA stack, their deployment status, and their cryptographic action provenance. The right-hand column is the relevant column for the argument of this paper.

Platform	VLA / ER Stack	Deployment Status (April 2026)	Action Provenance
Agility Digit	Proprietary (Agility)	100,000+ totes at GXO; contracts with Toyota, Mercado Libre	None
Apptроник Apollo	Gemini Robotics (Google)	Mercedes-Benz Germany & Hungary; \$350M Series A	None
Boston Dynamics Atlas	Gemini Robotics (DeepMind)	Post-CES 2026 production ramp; Hyundai committed	None
Figure 03	Helix 02 (in-house)	BMW + unnamed logistics; home pilots H2 2026	None
1X NEO	Redwood (in-house)	Consumer pre-orders open; 2026 home delivery	None
Tesla Optimus	xAI Grok	Limited internal Tesla facility use	None
XPENG IRON	XPENG Physical AI	Q1 2026 mass production launch	None
Unitree G1 / H1	Open-source + partner VLAs	5,500 shipped 2025; target 10–20k 2026	None
FANUC / KUKA / UR	Intrinsic Flowstate + Gemini	Broad automotive & electronics; Foxconn JV	None
da Vinci Surgical	Classical teleop + AI assist	Global hospital deployment	21 CFR Part 11 (pre-crypto)

The right-hand column is where the chapter coheres. Of the ten platforms listed, only surgical robotics operates under an audit-trail regime, and that regime — as §2 established and §5.4 develops further — is pre-cryptographic, centrally-logged, and designed for a deployment profile that does not match the 2026 distributed-edge-AI reality. The remaining nine platforms ship action logs to vendor-controlled infrastructure under standard retention policies with no cryptographic binding between the logs and the actions they purport to record. The physical-AI industry of April 2026 has assembled the technical capacity to deploy humanoids and intelligent industrial arms at scale without assembling the accountability infrastructure to certify the actions they produce.

5. The Regulatory Forcing Function

The industry state described in §4 would be unremarkable — a market in an early deployment phase, with accountability infrastructure to follow — if the regulatory framework had an equivalently early timeline. It does not. Three regulatory deadlines, all binding on operators of embodied-AI systems in the European Union, fall within a twelve-month window beginning August 2026. The corresponding United States frameworks, while less prescriptive, establish the same direction of travel. This section describes the deadlines and identifies the regulatory provisions that, in their plain reading, mandate propagation identity without naming it.

5.1 The Twelve-Month Deadline Stack

Effective Date	Instrument	Scope Relevant to Embodied AI
2 August 2026	EU AI Act, Annex III obligations	High-risk AI systems independent of regulated products: critical-infrastructure operation, employment decisions, biometric identification. Applies to standalone AI components of embodied systems.
20 January 2027	Machinery Regulation 2023/1230	Replaces Machinery Directive 2006/42/EC. Classifies AI software with safety functions as a safety component. Requires logs of all interactions to be maintained for traceability. Triggers re-conformity assessment when AI behaviour changes.
2 August 2027	EU AI Act, Annex I obligations	High-risk AI embedded in regulated products including machinery, medical devices, automotive, lifts, radio equipment. Aligns AI Act conformity with sector regulations including Regulation 2023/1230.

The three deadlines compose a cumulative obligation. An operator placing a Gemini-Robotics-driven humanoid into a German body shop in late 2026 is simultaneously subject to the AI Act Annex III obligations on the AI component, the Machinery Regulation obligations on the integrated system once January 2027 arrives, and the AI Act Annex I conformity assessment that aligns the two frameworks from August 2027. A compliance posture that addresses one deadline without addressing the others is structurally incomplete.

5.2 The Machinery Regulation Describes Propagation Identity in Regulatory Prose

Regulation (EU) 2023/1230 is the instrument that, on the closest reading, mandates the primitive defined in this paper. Three provisions are decisive.

Article 3, point 3 — definition of safety component. The Regulation defines a safety component as a component, including software, the failure or malfunction of which endangers the safety of persons. The plain reading of this definition, applied to a VLA policy whose runtime decisions determine the motion of a humanoid in a workplace, classifies the VLA policy as a safety component. A safety component is subject, under the Regulation, to conformity assessment by a notified body.

Annex III, Part B, paragraph 1.1.6 — traceability of interactions. The Regulation requires that, for machinery whose operation involves AI software with safety functions, ‘logs of all interactions need to be maintained for traceability.’ The text uses the term ‘interactions,’ not ‘operator inputs.’ It applies to interactions among layers of the system — including, on the plain reading, the interaction between the planner and the VLA policy, and between the VLA policy and the actuator. It thereby describes the propagation chain in regulatory prose.

Article 23 — re-conformity on AI behaviour change. The Regulation requires a new conformity assessment when the behaviour of the AI software changes. ‘Behaviour change’ is not given a strict numerical definition, but the practical implication is that the operator must be able to demonstrate which AI software was active at the time of any conformity assessment, and to detect when that software has changed. Without per-inference cryptographic attestation of the model in use, ‘behaviour change’ cannot be evidenced and therefore the re-conformity obligation cannot be discharged.

5.3 EU AI Act Provisions Relevant to Embodied AI

- Article 12 (record-keeping) requires high-risk AI systems to ensure the automatic recording of events over the lifetime of the system. The text does not specify cryptographic record-keeping; in the absence of such specification, the regulator’s ability to verify the integrity of the records is limited to the operator’s asserted log integrity.
- Article 26(6) (deployer obligations) requires deployers of high-risk AI to keep automatically generated logs for a period appropriate to the intended purpose of the system, and at minimum six months. The same observation about cryptographic integrity applies.
- Article 72 (post-market monitoring) requires providers of high-risk AI to establish post-market monitoring systems that document the performance of the AI throughout its lifetime. Performance documentation that is not cryptographically bound to the actions performed is, in adversarial conditions, indistinguishable from performance documentation that has been retroactively edited.

5.4 United States Frameworks

The United States regulatory picture is less prescriptive but consistent in direction. Executive Order 14110 (2023) explicitly calls for identity-verification standards for AI systems and content authentication. The NIST AI Risk Management Framework, and the related NIST work on AI identity, develops the provenance and accountability dimension as a core function. For surgical robotics, FDA 21 CFR Part 11 establishes audit-trail requirements that have governed the sector for two decades; these requirements predate cryptographic attestation and are designed for centralized hospital infrastructure. They are therefore the only existing US

precedent for action-level audit trails in physical machines, and they are not adequate for the distributed-edge-AI deployment profile of 2026 humanoid platforms.

5.5 Civil Liability and the Insurance Market

The revised Product Liability Directive ((EU) 2024/2853), in force since December 2024 and applicable from 9 December 2026, explicitly extends product-liability principles to AI systems and to software more generally. The Directive's evidentiary structure shifts a portion of the burden of proof toward the producer when the claimant cannot reasonably access internal information about the product. For an embodied-AI deployment, 'internal information' includes the chain of decisions that produced a harmful action; an operator without propagation identity is, under the new Directive, in a structurally weaker position to defend against a liability claim than an operator with it.

The insurance market is the second-order consequence. Underwriters of robotic operations are increasingly requiring evidentiary infrastructure that can be inspected post-incident. The forensic insurance market for cyber risks built itself around tamper-resistant log infrastructure during the period 2010–2020; the equivalent build-out for embodied-AI risks is now beginning, and the underwriting requirements will reflect the regulatory and product-liability environment described above.

5.6 The Forcing Function Summarized

The regulations are not asking for a new primitive to be invented. They are asking for the primitive that they have already implicitly mandated to be made operationally available before the deadlines that have already been set. Operators who do not possess the primitive will, beginning in August 2026, be placing systems on the market that they cannot demonstrate to be compliant. The window in which propagation identity can be developed in advance of the obligation has, at the date of this paper, narrowed to under sixteen months.

6. Existing Approaches and the Provenance Stack

6.1 The Six Layers as They Exist in 2026

The provenance and identity infrastructure that the computing industry has assembled over the past two decades can be decomposed, for the purposes of this paper, into five well-developed layers. The proposal of this paper is to add a sixth layer that composes the five and extends their reach across the boundary into physical actuation. The layers, numbered from the hardware base upward, are described in the table below.

Layer	Layer Name	Representative Primitives	Bound Object
L0	Hardware attestation	TPM 2.0; Intel TDX; AMD SEV-SNP; secure boot	Boot integrity; enclave state
L1	Media provenance	C2PA; Content Credentials	Media file (image, video, audio)
L2	Build-time provenance	SLSA; in-toto; Sigstore	Build artefact; release
L3	Workload identity	SPIFFE / SPIRE; W3C DID; Verifiable Credentials	Running workload; subject
L4	Agent action provenance	AARM; Google AP2 (Intent / Cart Mandates); Prova; Anjuna	Software action; payment; transaction
L5	Propagation identity (proposed)	This paper; GNS / GEIANT / MobyDB / ULissy	Physical action, bound to delegation, model, runtime, actuator, spatial cell, epoch

6.2 What Composition Means

Layer 5, as proposed, is not a replacement for any of the layers below it. It is the layer that composes them and projects the composition across the decision-to-action boundary. The composition is intended to be operationally concrete, not metaphorical.

A signed action emitted at Layer 5 inherits, by construction, attestations from each layer below. The actuator that produces the physical motion is a workload bound by Layer 3 to a runtime identity, running on a binary attested at Layer 2 by SLSA, on a hardware platform attested at Layer 0 by a TPM. The model whose inference produced the motor primitive is itself a workload bound by Layer 3, deriving from a binary attested at Layer 2, processing perceptions whose origin can in principle be attested by Layer 1 if the sensor pipeline is C2PA-aware. The software-agent action that triggered the inference, where applicable, is signed at Layer 4 by AARM or AP2. The Layer 5 signature is therefore a composite: it asserts not only that this physical action was produced by this signed chain, but that the chain itself rests on a verifiable foundation.

6.3 What Layer 5 Adds That Layers 0 Through 4 Cannot

Three properties of Layer 5 are not provided, individually or collectively, by Layers 0 through 4.

Spatial binding. No layer below Layer 5 binds an action to a verifiable spatial coordinate. The H3 hexagonal grid, used as the geospatial primary key in this proposal, gives every signed action a spatial address at hierarchical resolution. The address is part of the signed payload

and is therefore non-repudiable. The lower layers, designed for placeless software environments, have no equivalent.

Jurisdictional inheritance. The H3 cell, in combination with a published map from cells to legal jurisdictions, allows a signed action to inherit a regulatory framework. An action signed in cell 8a1f1d0b3c1 inherits the jurisdictional framework of the location of that cell. Lower layers do not have a primitive for this inheritance.

Delegation constraint propagation. Layer 5 signatures inherit constraints from upstream layers in a one-direction-only manner. A planner with delegated scope of ‘rack B-3’ cannot, even if it wished, produce a signed action at ‘rack B-2’ that the actuator’s verifier would accept. The constraint propagates downward through the chain by construction. SPIFFE workload identity supports scope but does not enforce it at actuation; AARM supports software-action constraints but does not bind to actuation. Layer 5 is the layer at which the constraint is enforced where it matters.

6.4 Positioning Statement

Every mature provenance primitive in 2026 solves part of the chain. Hardware attestation proves what booted. C2PA proves who observed. SLSA proves who built. SPIFFE proves who runs. AARM and AP2 prove what a software agent decided. Propagation identity is the primitive that connects them through the physical actuation. It does not displace the layers below; it is the layer those layers have been building toward.

The remainder of this paper develops the primitive in three phases. Part III defines it formally — the four-layer signing chain, the H3 spatial constraint, the epoch temporal anchor, the delegation inheritance rule. Part IV describes its implementation — the ULissy language construct propagates from, the MobyDB row schema, the GEIANT runtime enforcement model, and a worked example drawn from a Comau-style welding cell. Part V examines the implications — for insurance underwriters, for certification bodies, and for the standardization track that the GNS Foundation has begun with the Trajectory Identity Protocol Internet-Draft.

PART III

The Primitive

7. Formal Definition of Propagation Identity

7.1 Definition

Propagation identity is the property of an agentic physical system whereby every action in the physical world is cryptographically traceable back through the full chain of decisions that produced it — human intent, model inference, runtime planning, actuator command — with each layer signing its own contribution and inheriting constrained authority from the layer above.

The definition is operational: it specifies a property that a system either has or does not have, and that can be verified by inspection of any single signed action and the chain of certificates referenced from it. The definition is not a design pattern. It is a primitive of the same kind as TLS, code signing, or hardware attestation: a verifier presented with the artefacts produced by the system can decide, in finite time and without trusting the operator, whether the property holds.

7.2 The Three Non-Negotiable Properties

A primitive that satisfies the definition above necessarily exhibits three properties. Each is non-negotiable in the sense that a candidate primitive lacking any one of them does not solve the problem of §1 through §3.

P1 — Composability. The chain must span software-agent primitives (Layer 4 — AARM, AP2, Prova) and physical-actuation primitives (Layer 0 — TPM, TDX-rooted hardware) without translation loss. A signature emitted by an AARM-class agent must be verifiable by the same verifier that checks the actuator’s hardware attestation. The primitive does not introduce new trust roots; it composes existing ones.

P2 — Spatial binding. Each signed action is anchored to a verifiable spatial coordinate, expressed as an H3 hexagonal cell at a specified resolution, and to a sealed temporal epoch. The (cell, epoch) pair is part of the signed payload and is therefore non-repudiable. Spatial binding is what makes jurisdictional inheritance possible; without it, the primitive devolves into Layer 4.

P3 — Delegation constraint. Authority flows downward through the chain and is narrowable, never expandable. A downstream layer cannot, even with valid signing keys, produce an action whose scope exceeds the scope granted by the upstream certificate. The constraint is enforced at the verifier, not at the signer; an actuator that signs an out-of-scope action produces a signature that no compliant verifier will accept.

7.3 Notation Conventions

The remainder of this paper uses the following notation. A signing key pair belonging to layer L is written (sk_L, pk_L) ; only the public component pk_L is written when only the verification side is relevant. A signature by layer L over message m is written $sig_L(m)$. The H3 cell of an action is written $cell_R(p)$ where R is the resolution and p is the position; resolution 15 (approximately 0.9 m^2) is used as the default unless otherwise noted. A sealed epoch is written e , with Merkle root $M(e)$. The composite primary key on which propagation records are stored is written $(cell, e, pk_{actor})$.

8. The Four-Layer Signing Chain

8.1 Structural Overview

Propagation identity is realized as a four-signature breadcrumb. Each signature is produced by the public key of one layer in the chain, over a payload that includes a hash of the upstream certificate and a hash of the downstream contribution. The four signatures, taken together, constitute the proof that the action was authorized, decided, planned, and actuated by the named identities.

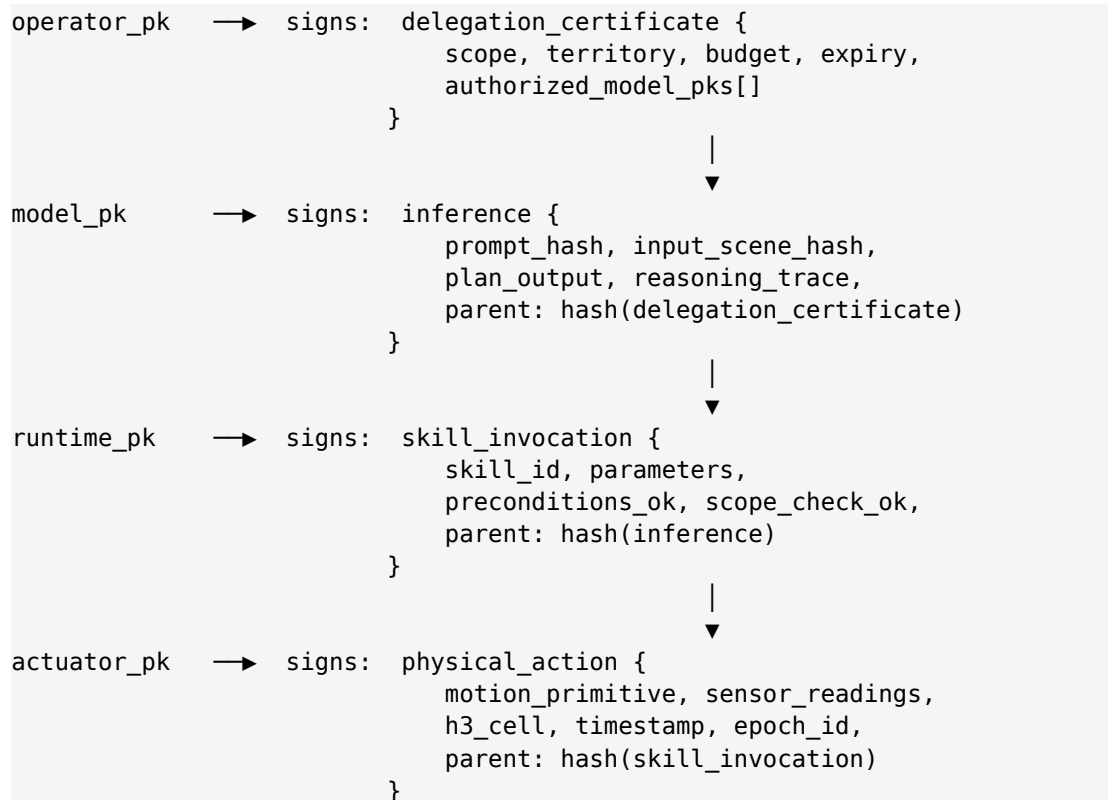


Figure 1 — The four-layer signing chain. Each layer's payload includes a hash of the upstream certificate.

8.2 Layer Responsibilities

- The operator layer issues a delegation certificate that constrains scope (which tasks), territory (which H3 cells), budget (how many actions or how much energy), expiry (until when), and the set of authorized model public keys. The certificate is signed by the operator's key, which is itself rooted in a human GNS identity with Proof-of-Trajectory.
- The model layer signs the inference. The signed payload includes the hash of the prompt, the hash of the perceived input scene, the structured plan output, and the natural-language reasoning trace where one is produced by an embodied-reasoning model such as Gemini Robotics ER. The signature binds this specific model run to this specific delegation.
- The runtime layer signs the skill invocation. The runtime is responsible for verifying that the model output corresponds to a known skill, that the parameters are within bounds, and that preconditions and scope checks have passed. The runtime signature is the layer at which the delegation constraint is enforced; a runtime that signs an invocation outside scope produces a signature that the actuator's verifier will reject.
- The actuator layer signs the physical action. The actuator records the actual motion primitive issued, the sensor readings observed during execution, the H3 cell at the moment of actuation, the wall-clock timestamp, and the current open epoch identifier. The actuator's signing key is rooted in hardware attestation (TPM 2.0 or equivalent).

8.3 Size, Latency, and Throughput

Each breadcrumb consists of four Ed25519 signatures (256 bytes total), the payloads (roughly 800 to 1,200 bytes depending on the verbosity of the reasoning trace), and the spatial-temporal anchor (96 bytes for H3 cell, epoch identifier, and timestamp). A typical breadcrumb is therefore between 1.4 and 1.8 kilobytes.

Signing rate	Layer at which signed	Per-robot bandwidth	Daily volume
1 kHz	Joint command (not recommended)	~5 MB/s	~430 GB/day
100 Hz	Skill-step command	~500 KB/s	~43 GB/day
10 Hz	Skill invocation (recommended)	~50 KB/s	~4.3 GB/day
~1 Hz	Task completion	~5 KB/s	~430 MB/day

The recommended deployment configuration signs at the skill-invocation rate (approximately 10 Hz) rather than at the joint-command rate. Skill-invocation signing produces a verifiable record of every meaningful decision the system makes — every grasp, every place, every weld, every traversal — at a bandwidth that is unproblematic on commodity edge hardware

and at a daily volume that is straightforward to retain for the six-month minimum required by EU AI Act Article 26(6) and the longer windows that may be required by the Machinery Regulation. Ed25519 signing on a modern ARM core completes in under 100 μ s, well below the latency budget of a 10 Hz control loop.

9. Spatial Constraint via H3 and the GeoEpoch Protocol

9.1 Why a Geospatial Primitive Is Required

A propagation chain that does not include a verifiable spatial coordinate cannot inherit a regulatory framework, because the framework is selected by location: an action signed in Hungary is governed by Hungarian and EU law; the same action signed in Texas is governed by Texas and US federal law. Without spatial binding, the primitive devolves into a Layer 4 software-agent provenance scheme. The GeoEpoch Protocol (GEP), defined in earlier work by the GNS Foundation and used as the spatial substrate of the GNS Protocol, provides the binding.

9.2 H3 as Primary Key

The H3 hexagonal hierarchical grid, originally developed at Uber and now maintained as an open-source standard, partitions the surface of the Earth into hexagonal cells at sixteen resolutions. Resolution 15, the finest, divides the Earth into approximately 569.7 trillion cells with an average area of 0.9 m². Resolution 9 produces cells of roughly 105,000 m², appropriate for warehouse-zone or factory-line authorization. Resolution 7 produces cells of 5.16 km², appropriate for jurisdictional authorization at the city or region scale. The hierarchical containment property — every cell at resolution R is contained in exactly one cell at resolution R-1 — makes spatial range queries native and makes delegation at one resolution enforceable at any finer resolution.

In the propagation primitive, the actuator records, at the moment of physical motion, the H3 cell of the end-effector or centre of mass at resolution 15. The recorded cell is part of the signed payload. A verifier checking compliance against a delegation certificate that authorizes operation in a region defined by a set of resolution-9 cells need only check that the resolution-15 cell of the action is contained, by H3's native containment relation, in one of the authorized resolution-9 cells. The check is constant-time and does not require external geographic services.

9.3 GEP as the Anchor

The **GeoEpoch Protocol** establishes a public-domain genesis hash that fixes the H3 grid against a specific Earth datum and a specific epoch (the protocol's own genesis time). The GEP genesis hash, 26acb5d998b63d54f2ed92851c5c565db9fe0930fc06b06091d05c0ce4ff8289, is referenced by every signed action in the propagation primitive. The reference makes the

spatial coordinate system itself non-repudiable: a verifier who knows the genesis hash and the H3 specification can reconstruct, exactly, what region of the physical world the recorded cell refers to. There is no version-skew between the operator's map and the actuator's map.

9.4 Jurisdictional Inheritance

A separate, public, signed map associates H3 cells with jurisdictional frameworks. The map is published once and updated only at boundary changes (which are infrequent at the resolutions relevant to embodied AI). A signed action at cell c inherits, by lookup, the legal framework applicable at c . The inheritance is local to the verifier; the signer does not need to know the framework when signing. This separation is structurally important, because it means that operators in different jurisdictions can use the same identical propagation infrastructure and have the verifier produce different compliance outputs.

10. Temporal Anchoring via Epoch Sealing

10.1 Breadcrumbs and Epochs

Individual signed actions, in the propagation primitive, are termed breadcrumbs — a term inherited from the GNS Protocol, where the same construct is used for human Proof-of-Trajectory. Each breadcrumb is independently signed and can be independently verified. Breadcrumbs accumulate during normal operation; periodically, the set of breadcrumbs that has accumulated since the last sealing event is sealed into an epoch.

An epoch is a record of the form $(e_id, M(e), t_open, t_close, signer_pk)$, where e_id is a unique identifier, $M(e)$ is the Merkle root of the breadcrumbs sealed in the epoch, t_open and t_close are the timestamps that bound the epoch's breadcrumbs, and $signer_pk$ is the public key of the entity sealing the epoch. The epoch record is itself signed and broadcast to the GNS network, where it is anchored by inclusion in the next Stellar mainnet block reference.

10.2 What Sealing Provides

Once an epoch is sealed, the set of breadcrumbs it contains becomes immutable in a strong sense: a verifier presented with any single breadcrumb and the corresponding Merkle inclusion proof can confirm, against the sealed epoch record, that the breadcrumb was indeed part of the epoch and was not added after the fact. The operator cannot retroactively edit a single signed action without invalidating the Merkle inclusion proof of every other breadcrumb in the same epoch. This property is what distinguishes propagation identity from log-based audit trails: a log can be edited; a Merkle-rooted epoch cannot.

10.3 Epoch Cadence

Operational cadence for sealing matches the rhythm of the deployment. A factory floor operating in three shifts may seal four epochs per day (one per shift, plus a closing epoch). A continuous logistics deployment may seal one epoch per hour. A surgical-robotics deployment

may seal one epoch per procedure. The cadence is configurable per deployment; the only constraint is that the cadence is published as part of the operator's compliance posture and must match what the operator subsequently asserts in regulatory filings.

11. Storage and the Composite Primary Key

11.1 The (cell, epoch, public_key) Triple

Propagation breadcrumbs are stored under a composite primary key consisting of the H3 cell of the action, the epoch identifier of the sealing event in which the breadcrumb is included, and the public key of the actor producing the action. The triple is naturally unique: at any single H3 cell, in any single epoch, an actor with a given public key produces at most one breadcrumb of a given type. Composite indexing on the triple supports the three query patterns that compliance and forensics workloads require: spatial range queries (give me all actions in this region during this audit window), temporal range queries (give me all actions during this incident window), and identity queries (give me all actions by this specific agent).

11.2 MobyDB-Native Storage

MobyDB, the geospatial-native database developed alongside the GNS Protocol, treats the H3 cell as the primary key by construction. Its 48-byte composite key (Ed25519 identity + H3 cell + Stellar wallet) is the same composite key that propagation breadcrumbs naturally produce. Storing propagation records in MobyDB is not an integration; it is a coincidence of design. The Inference and Imagery collection types defined in MobyDB's base schema accommodate, respectively, the model-layer payload and any sensor frames the actuator chooses to attest at Layer 1 via C2PA.

11.3 Retention and Cost

At the recommended skill-invocation signing rate, a single robot produces approximately 4.3 GB of breadcrumb data per day. Six months of retention, the minimum required by EU AI Act Article 26(6), corresponds to approximately 800 GB per robot. At commodity object-storage prices (USD 0.005 per GB-month) the storage cost is roughly four US dollars per robot per month. The cost is unproblematic relative to the operational cost of the robot itself. The retention horizon is a regulatory parameter, not a technical constraint.

PART IV

Implementation

12. ULissy Grammar: the propagates from Construct

12.1 Motivation

The propagation primitive can in principle be implemented as a runtime library: a robotics-control framework that wraps motor commands with signing logic and emits breadcrumbs to a persistent store. The library approach is workable but fragile. A developer can forget to wrap a command, or wrap it incorrectly, or pass the wrong upstream certificate, and the omission will not be caught until an auditor or a regulator examines a record that should exist and does not.

The ULissy domain-specific language, defined in the ULissy v0.1 Whitepaper, treats identity, location, time, cryptography and energy as first-class language primitives. Adding propagation identity as a language construct rather than a library call replaces a runtime convention with a compile-time guarantee. A program that compiles to a propagating action is a program in which the chain of certificates was statically verified to exist; a program that does not compile is a program that the developer cannot ship.

12.2 Syntax

The propagates from construct binds an action expression to its upstream chain. The chain is an explicit set of (layer, public_key, certificate_reference) tuples; the compiler verifies that each tuple has a corresponding certificate in scope and that the scopes are consistent.

```
// A welding skill invocation expressed in ULissy

skill weld_seam(
  seam: Seam,
  fixture: H3Cell,
  parameters: WeldParameters
) -> Result<WeldRecord>
  propagates from {
    operator: @comau-cell-7,           // delegation cert
    model:    gemini.robotics.er@1.6,  // signed inference
    runtime:  flowstate.runtime@2.4,   // signed invocation
  }
  within fixture                       // spatial constraint
  budget 30.seconds                   // temporal budget
  signs_at actuator.tpm               // hardware-rooted key
{
  let pre = check_preconditions(seam, fixture)?;
  let trajectory = plan_trajectory(seam, parameters)?;

  for primitive in trajectory {
    actuate(primitive)                // each primitive emits
                                     // a signed breadcrumb
  }
}
```

```

    seal_breadcrumb(seam, fixture, parameters)
}

```

Figure 2 — A welding skill expressed with the ULissy propagates from construct. The compile-time signature chain is explicit; the runtime signing happens automatically.

12.3 Compile-Time Guarantees

The ULissy compiler enforces three properties at compile time. First, every action expression annotated with propagates from must reference a certificate from each of the operator, model, and runtime layers; a missing layer is a compile error, not a runtime warning. Second, the within clause must be a subset of the operator certificate's territory; a skill that could in principle execute outside the delegated territory will not compile. Third, the budget clause must be inside the operator certificate's budget; a skill that could consume more than the delegated budget will not compile. The runtime layer's job is reduced from policy enforcement to certificate freshness checks.

13. MobyDB Row Structure for Propagation Records

Propagation breadcrumbs are stored as rows in a MobyDB Inference collection. The schema below is illustrative; actual deployments may add operator-specific fields under the metadata column.

```

CREATE COLLECTION propagation_breadcrumbs (
  h3_cell      H3Cell    NOT NULL,      -- spatial primary
  epoch_id     Epoch     NOT NULL,      -- temporal primary
  actor_pk     Ed25519   NOT NULL,      -- identity primary
  timestamp    Timestamp NOT NULL,

  -- Action payload
  action_type  Enum      NOT NULL,      -- skill | step | task
  skill_id     Text,
  parameters   JSON,
  sensor_digest Hash,                  -- content hash (C2PA)

  -- Chain references (hashes of upstream payloads)
  parent_chain Hash[4]   NOT NULL,      -- 0 / M / R / A

  -- Signatures — one per layer
  sig_operator Ed25519Sig NOT NULL,
  sig_model    Ed25519Sig NOT NULL,
  sig_runtime  Ed25519Sig NOT NULL,
  sig_actuator Ed25519Sig NOT NULL,

  -- Composability hooks
  tpm_quote    Bytes,                  -- Layer 0 attestation
  spiffe_id    Text,                   -- Layer 3 workload id
  aarm_ref     UUID,                   -- Layer 4 ref (if any)
)

```

```

metadata      JSON,

PRIMARY KEY (h3_cell, epoch_id, actor_pk, timestamp)
);

```

Figure 3 — MobyDB collection schema for propagation breadcrumbs. The composite key matches MobyDB's native 48-byte (cell, epoch, identity) layout.

Three properties of this schema are worth noting. The `parent_chain` column carries the hashes of the four upstream payloads (operator delegation, model inference, runtime invocation, actuator action), permitting any verifier to walk the chain. The composability hooks (`tpm_quote`, `spiffe_id`, `aarm_ref`) allow the breadcrumb to reference Layer 0, Layer 3, and Layer 4 attestations respectively, without embedding their full payloads — the breadcrumb is a pointer into the existing provenance stack rather than a duplication of it. The PRIMARY KEY is exactly the (cell, epoch, identity) triple defined in §11.

14. GEIANT Runtime Enforcement: TierGate for Physical Agents

14.1 Existing TierGate

The GNS Protocol's TierGate component scores software agents on a five-tier scale (Provisioned → Observed → Trusted → Certified → Sovereign) based on accumulated compliant operations within a declared jurisdiction. TierGate is the runtime layer at which a software agent's capability is gated by its observed compliance history. The same mechanism extends naturally to physical agents under propagation identity, with one extension.

14.2 Extension for Physical Agents

A physical-agent tier is computed not only from operation count and compliance history, but also from three additional dimensions visible only when the chain is signed.

- Territorial fidelity — the proportion of actions whose recorded H3 cell is within the operator's authorized territory. A high-fidelity agent rarely strays; a low-fidelity agent has produced multiple breadcrumbs at cells outside scope, even if those actions were ultimately corrected.
- Delegation discipline — the proportion of skill invocations whose runtime `scope_check_ok` flag was true at signing time. A disciplined agent is one whose runtime never had to refuse an out-of-scope request because no out-of-scope requests were produced upstream.
- Chain integrity — the proportion of breadcrumbs in which all four signatures verified, all four parent hashes resolved, and the epoch sealing closed cleanly. A chain-integral agent produces audit-ready records by default.

The tier is recomputed on each epoch sealing event. An agent that drops below threshold on any one of the three dimensions is automatically restricted to a lower tier until remediation; an agent that exceeds threshold on all three for a sustained period is promoted. The tier itself is a

signed attestation by the GEIANT runtime and is consumable by external systems — for example, by a customer placing orders only with Tier-Certified or higher cells.

15. Worked Example: a Comau-Style Welding Cell

15.1 Setup

Comau, the Italian automation company that was Intrinsic's first industry partner in 2023, produces modular automotive workcells. The example below describes a hypothetical Comau workcell deployed at a German automotive plant in early 2027. The cell consists of two six-axis articulated arms, a positioning fixture, and a perception subsystem comprising stereo cameras and a structured-light scanner. The cell is operated under Intrinsic Flowstate, with skills supplied by both Comau (welding) and the OEM (panel handling). The AI component is Gemini Robotics ER 1.6 for planning, with the lower-level VLA policy executing on the workcell's edge compute.

The cell occupies an area corresponding to four resolution-12 H3 cells (each approximately 240 m²). The operator's delegation certificate authorizes welding operations within these four cells, between 06:00 and 22:00 local time, for a budget of up to 480 weld seams per shift, on SKU 1842 only.

15.2 A Single Action

A welding action begins when the operator schedules a unit of SKU 1842 onto the cell. The schedule is itself a signed delegation refinement that references the cell's standing certificate. The planner — Gemini Robotics ER 1.6 — receives the SKU specification and produces a reasoning trace, a sequence of motor primitives for both arms, and an explicit grasp-and-place plan for the panel handler. The planner's output is signed by the model layer's key (rooted in a key-attested provisioning certificate from Intrinsic).

The runtime — the Flowstate execution engine extended with the propagates from compile-time guarantees — verifies that the plan corresponds to known skills (`weld_seam`, `place_panel`), that the parameters are within the cell's envelope, and that the predicted final position of each motion is within one of the four authorized H3 cells. The runtime signs the invocation.

The actuator — the lower-level controller of each arm — executes the motion primitives and, at completion of each weld seam, records the actual H3 cell of the seam endpoint, the sensor readings during the weld (current, voltage, infrared profile), the wall-clock timestamp, and the open epoch identifier. The actuator signs the breadcrumb with a key rooted in the controller's TPM 2.0.

15.3 The Resulting Breadcrumb

```
{
  "h3_cell":      "8c1f1d0b3c123ff",      // res-12 cell of seam end
```

```

"epoch_id":      "comau-cell-7-2027-04-30-shift-1",
"actor_pk":      "ed25519:6a9c...",           // arm-1 TPM-rooted
"timestamp":     "2027-04-30T08:14:22.117Z",

"action_type":   "skill",
"skill_id":      "weld_seam",
"parameters":    { "seam_id": "L-pillar-inner", "current_A": 220 },
"sensor_digest": "blake3:f1c8...",           // C2PA-attested

"parent_chain": [
  "sha256:operator-delegation-2027-04-30...",
  "sha256:geminier-1.6-inference-...",
  "sha256:flowstate-runtime-invocation-...",
  "sha256:arm1-controller-action-..."
],

"sig_operator":  "ed25519:9f44...",
"sig_model":     "ed25519:c2ab...",
"sig_runtime":   "ed25519:7e30...",
"sig_actuator":  "ed25519:b18d...",

"tpm_quote":     "bytes:6f01...",           // Layer 0
"spiffe_id":     "spiffe://comau/cell-7/arm-1",
"aarm_ref":      null                       // no software agent
}

```

Figure 4 — A single welding-action breadcrumb. ~1.6 KB on the wire.

15.4 Audit Export

At the close of an audit window — for example, a quarterly internal compliance review or a notified-body inspection under the Machinery Regulation — the cell exports a signed audit package. The package contains the set of sealed epochs covering the window, the Merkle inclusion proofs for the requested breadcrumbs, the certificates of every key referenced in the chain, and a compliance summary cross-referenced to the relevant articles of the AI Act and the Machinery Regulation.

The auditor verifies the package offline, against the GEP genesis hash and the public certificate authorities of the involved parties. The verification is constant-time per breadcrumb, requires no contact with the operator, and produces a deterministic answer: every action in the audit window was, or was not, the product of an unbroken chain rooted in a delegation that the operator can produce on demand.

PART V

Implications

16. Insurance and Civil Liability

16.1 The Cyber-Insurance Precedent

Between 2010 and 2020 the cyber-insurance market matured around a small number of evidentiary infrastructure requirements. Underwriters of cyber risk increasingly demanded tamper-resistant log infrastructure, signed configuration baselines, and verifiable incident-response artefacts before issuing coverage at standard rates. The infrastructure was not initially mandated by regulation; it was selected by the underwriting market because, in the absence of it, claims could not be adjudicated. The insured operators built the infrastructure to obtain coverage; the regulators eventually codified what the market had already required.

The same trajectory is plausible for embodied-AI risk. Underwriters of physical-AI deployments — Munich Re, Swiss Re, Allianz Trade and the specialist robotics underwriters that have entered the market since 2024 — face an adjudication problem identical to the early cyber market. When a humanoid causes a worker injury, when an industrial arm produces a non-conforming part that enters the supply chain, when an autonomous logistics unit damages adjacent equipment, the claim cannot be adjudicated without a forensically reliable record of what the system did and on whose authority. Unsigned vendor logs are the equivalent of self-reported activity logs in the early cyber market: legally admissible, evidentially weak, commercially insufficient.

16.2 The Product Liability Directive 2024/2853

The revised Product Liability Directive, applicable from 9 December 2026, establishes that, where a claimant is unable to prove a defect or causal link due to technical or scientific complexity, the burden shifts toward the producer. The Directive specifically anticipates AI systems and software in this provision. An operator who can produce a complete propagation chain for the relevant action is in a structurally strong evidentiary position; an operator who cannot is in a structurally weak position, exposed to the burden-shifting regime that the Directive creates. The Directive does not name propagation identity. It creates the evidentiary asymmetry that makes propagation identity commercially valuable.

16.3 The First-Mover Advantage in Underwriting

The first underwriter to require, or to discount premiums for, deployments that produce propagation chains will gain the same competitive advantage that the first cyber underwriters obtained by requiring tamper-resistant logging in the 2012–2015 period. The advantage is structural: the underwriter that requires the artefact obtains, at no marginal cost, a higher-quality claims-adjudication process than competitors that do not require it. The market signal travels from underwriter to broker to operator, and the infrastructure becomes a precondition of cover.

17. Certification Frameworks: The Notified-Body Bridge

17.1 The Notified-Body Model

Conformity assessment under the Machinery Regulation 2023/1230, for machines incorporating safety-related AI software, is performed by an EU-accredited notified body. The major notified bodies in the field include TÜV SÜD, TÜV Rheinland, Bureau Veritas, DEKRA and INERIS. The notified body inspects the design, the production process, and — critically for the argument of this paper — the operator’s demonstration that the AI software conforms to the safety requirements set out in Annex III. The output of the assessment is a CE certification that authorizes placing the machinery on the EU market.

In the regime that prevails up to 19 January 2027, AI software is assessed against generic safety-component criteria. Beginning 20 January 2027 the Regulation specifies, in its plain reading, that the operator must produce traceability of all interactions among system components. The operator who can produce a propagation chain has, in the same artefact, satisfied multiple distinct conformity sub-requirements: traceability under Annex III, evidence of competent design under Article 11, and evidence of post-deployment behaviour stability under Article 23.

17.2 A Standardization Opportunity

The notified bodies are presently developing internal methodologies for assessing AI safety components in the absence of a sectoral cryptographic-attestation standard. Each notified body’s methodology is, today, an internal document. The opportunity for the GNS Foundation, in cooperation with operators (Comau, Apatronik, Boston Dynamics, Universal Robots) and with the notified bodies themselves, is to propose a sectoral schema that the notified bodies can adopt as a common reference. The schema would specify the structure of the audit-export package (§15.4), the minimum-acceptable signing rate, the retention horizon, and the verification procedure. A schema agreed by two or more major notified bodies becomes, in practical terms, a de-facto standard for the sector.

17.3 Bridge from “Logs” to “Signed Chains”

The transition from log-based audit to chain-based audit is not, for any individual operator, a single decision. It is a sequence of intermediate states, each of which raises the floor of what an auditor can be expected to verify.

- State 0 — Vendor logs only. Audit relies on the operator’s assertion of log integrity. Current default for nine of the ten platforms surveyed in §4.
- State 1 — Signed log batches. Operator periodically signs log files with a deployment key. Tamper-evident at the file level, not at the action level. Improvement over State

0 but does not satisfy the Machinery Regulation’s per-interaction traceability requirement.

- State 2 — Signed actions, single layer. Each action is signed by the actuator. Sufficient for proving that the action occurred; insufficient for proving on whose authority.
- State 3 — Full propagation chain. Each action is signed by all four layers; chain is verifiable end-to-end. Satisfies the Machinery Regulation, the AI Act record-keeping articles, and the Product Liability Directive evidentiary structure simultaneously.

The argument of this paper is that State 3 will be the operating regime within the period 2027–2029, and that operators who plan their compliance posture against State 0 or State 1 will incur a transition cost during that period. Operators who deploy directly into State 3 incur no transition cost.

18. Open Questions

Six questions remain open at the publication of this draft. Each is identified here for community input ahead of the v1.0 of this paper.

18.1 Key Rotation Under Propagation

The signing keys at each layer of the chain — operator, model, runtime, actuator — must rotate over their operational life: operator keys when personnel change, model keys when models are re-released, runtime keys when systems are re-imaged, actuator keys when hardware is serviced. Rotation must not invalidate historical breadcrumbs. The natural mechanism is a rotation certificate signed by both the outgoing and incoming keys, included by reference in subsequent breadcrumbs. The detailed schema and revocation semantics are specified in v1.0.

18.2 Adversarial Model Substitution

The model layer’s signing key is rooted in a key-attested provisioning certificate. An adversary capable of compromising the model provider’s provisioning infrastructure could in principle substitute a malicious model whose signatures are nominally valid. The defence is multi-party signing of provisioning certificates and continuous behavioural attestation against a published reference dataset. The adversarial-substitution analysis is referenced from but not exhausted by this paper.

18.3 Cross-Jurisdiction Actions

An action that begins in one H3 cell and ends in another may cross a jurisdictional boundary mid-execution. The naive convention — assign the action to the cell of its endpoint — is insufficient for, e.g., an autonomous vehicle traversing the German-Polish border with an action whose execution spans the boundary. The proposed convention is to record the cell at

the moment of decision (model inference) and the cell at the moment of actuation, and to inherit the more restrictive of the two jurisdictions when they differ. The convention is sufficient for most cases; the unusual cases are enumerated in v1.0.

18.4 Long-Tail Actuators Without Hardware Attestation

A material fraction of installed industrial equipment does not carry TPM 2.0 or equivalent hardware attestation. The deployment cost of retrofitting is high. Two paths are available. The first is software-only attestation rooted in a network-provisioned identity, accepting a weaker security posture in exchange for deployment feasibility; this path is acceptable for low-criticality equipment but inadequate for safety-critical machinery. The second is a software-attested gateway model, in which an attested edge controller signs on behalf of an unattested actuator, with an explicit declaration in the breadcrumb that the actuator-layer signature is by-proxy. Both paths are described in v1.0.

18.5 Model Identity Versus Model Weights

When a model is fine-tuned, the resulting model is, in some senses, a new model and, in some senses, the same model. The propagation primitive requires that the model layer's identity is bound to the specific weights that produced the inference. The mechanism is a content hash of the weights (Blake3 or SHA-256), included in the model-layer payload. Fine-tuning produces a new content hash and therefore a new model identity; the relationship between the parent and child models is recorded in a separate provenance graph but is not part of the propagation chain.

18.6 Privacy and Spatial Trajectories

A complete propagation record contains the H3 cell of every action a robot takes. At resolution 15, this is a movement trace at sub-metre granularity. For deployments in environments where the trace is itself sensitive — for instance, a humanoid operating in a private home — the operator may need to publish only coarser-resolution cells in the audit-export package while retaining finer-resolution records internally. H3's hierarchical containment supports this: the public package contains breadcrumbs at resolution 9 or 10 while the operator's internal records retain resolution 15. The redaction is operator policy, not a property of the primitive.

19. Call to Standardization

19.1 The IETF Path

The GNS Foundation has established a standardization track at the IETF with the Trajectory Identity Protocol Internet-Draft draft-ayerbe-trip-protocol-03, which defines Proof-of-Trajectory for human identity. Propagation identity is the natural extension of TrIP to physical-AI agents and the corresponding Proof-of-Jurisdiction. The recommended standardization path is a follow-on Internet-Draft, draft-ayerbe-prop-identity-00, defining the four-layer signing chain,

the H3-anchored payload structure, the epoch sealing semantics, and the audit-export schema. The draft is intended for discussion in the IETF SECDISPATCH working group and, if scoped narrowly enough, in a future AI Cryptographic Engineering working group.

19.2 Coordination With Other Standards Bodies

- W3C Verifiable Credentials Working Group — coordination on the delegation-certificate format. The existing VC data model is suitable for the operator-layer certificate; the propagation chain is a domain-specific extension.
- OASIS C2PA — coordination on the perception-attestation hook. The propagation chain references C2PA-attested sensor frames at the actuator layer; the reference must be schema-compatible with C2PA.
- ISO/IEC JTC 1/SC 42 (Artificial Intelligence) — coordination on AI governance terminology and on the trustworthy-AI framework. Propagation identity is an evidentiary mechanism for properties that SC 42 defines abstractly.
- CEN-CENELEC JTC 21 — coordination on the harmonised standards that will accompany the AI Act. Propagation identity is candidate evidence for several of the harmonised-standard requirements under development as of April 2026.
- EU AI Office — coordination on the implementation guidance for AI Act Article 12 and Article 26(6). The Office has signalled interest in cryptographic-attestation approaches; propagation identity is one such approach.
- NIST AI Identity and Governance — coordination on the evolving NIST framework. Propagation identity is consistent with the NIST AI RMF function categories and adds a verifiable-evidence layer.

19.3 Patent and Open-Source Posture

USPTO Provisional Patent Application #63/948,788, held by the GNS Foundation, covers the Proof-of-Trajectory innovation that is the parent invention of propagation identity. The Foundation's declared patent posture, in line with its Commercial Open Source Software model, is non-assertion against open-source implementations of the protocol and licensing on FRAND terms to commercial implementations. The intent is to maximize adoption while retaining the option of defensive assertion against bad-faith forks.

19.4 Closing

The window in which propagation identity can be developed in advance of the regulatory obligation that will require it has narrowed, at the date of this paper, to under sixteen months. The primitive is constructible from existing components — Ed25519, H3, Merkle trees, TPM 2.0, the GNS Protocol, MobyDB, the ULissy language. The regulatory framework that will require it is already drafted and signed into law. The industrial deployments that will be subject to the framework are already operating. The notified bodies that will assess conformity are already convening their methodologies.

The remaining task is the engineering one: to define the schemas, to produce the reference implementations, to seek standardization through the established bodies, and to do so before the deadlines force less-considered solutions to fill the gap. The argument of this paper is that

the work can be done, that it should be done, and that the cost of not doing it — in regulatory exposure, in liability exposure, and in the structural opacity of physical AI to its principals and to its bystanders — is substantially higher than the cost of doing it.

APPENDICES

Glossary, References, Document Information

Appendix A — Glossary of Terms

AARM. Agent Action Reasoning Model. A 2026 framework for cryptographic provenance of software-agent actions, providing Layer 4 attestation in the stack of §6.

AP2. Google Agent Payments Protocol. Extension to A2A and MCP defining Intent and Cart Mandates as signed delegation artefacts for commerce-bound software agents.

Breadcrumb. A single signed action record in the propagation primitive. Contains four signatures (operator, model, runtime, actuator), an H3 cell, an epoch identifier, and a payload of approximately 1.4–1.8 KB.

C2PA. Coalition for Content Provenance and Authenticity. Industry standard for cryptographic provenance of media files; provides Layer 1 attestation in the stack of §6.

Delegation certificate. An operator-signed artefact specifying the scope, territory, budget, expiry and authorized model public keys for a deployment. The root of the propagation chain.

Ed25519. Edwards-curve Digital Signature Algorithm over Curve25519. The signing scheme used at every layer of the propagation primitive. Specified in RFC 8032.

Epoch. A sealed Merkle-rooted set of breadcrumbs covering a published time interval. The unit of immutability in the propagation primitive.

GEIANT. The geospatial-AI orchestration runtime developed by ULISSY S.r.l. The runtime layer at which TierGate scoring is computed and at which the propagation primitive is enforced for deployed systems.

GeoEpoch Protocol (GEP). Layer-0 geographic addressing fabric for the GNS Protocol, built on H3, with a public-domain genesis hash binding the H3 grid to a specific Earth datum.

GNS. Geospatial Naming System. The decentralized human-identity protocol on which the propagation primitive's human-side identities are rooted.

GNS-AIP. GNS AI Identity Protocol. The extension of GNS to AI agents, providing Proof-of-Jurisdiction. Propagation identity is the physical-AI specialization of GNS-AIP.

H3. Hexagonal hierarchical geospatial indexing system, originally developed at Uber. Sixteen resolutions; resolution 15 cells average 0.9 m². Used as the primary key for spatial binding in the propagation primitive.

Layer 5. The proposed layer of the provenance stack defined by this paper, composing Layers 0 through 4 and projecting their attestations across the decision-to-action boundary.

MobyDB. Geospatial-native database built on the GeoEpoch Protocol, with H3 cell as primary key. Native storage substrate for propagation breadcrumbs.

Notified body. An EU-accredited certification organization authorized to perform conformity assessment under the Machinery Regulation 2023/1230. Major bodies include TÜV SÜD, TÜV Rheinland, Bureau Veritas, DEKRA and INERIS.

Propagation identity. The property of an agentic physical system whereby every action in the physical world is cryptographically traceable back through the full chain of decisions that produced it. Defined formally in §7.

Proof-of-Jurisdiction. The trust-scoring property of an AI agent whose operations within its declared territorial scope are auditable. The AI-agent counterpart to Proof-of-Trajectory.

Proof-of-Trajectory. The property of a human GNS identity whose movements are recorded as cryptographically signed breadcrumbs. The human-identity primitive of GNS.

SLSA / in-toto. Supply-chain Levels for Software Artifacts; in-toto. Frameworks for build-time provenance of software artefacts. Layer 2 in the provenance stack.

SPIFFE / SPIRE. Secure Production Identity Framework for Everyone; SPIFFE Runtime Environment. Standard for workload identity in distributed systems. Layer 3 in the provenance stack.

TierGate. The GNS Protocol component that scores agents on a five-tier scale based on accumulated compliant operations. Extended in §14 for physical agents.

TPM. Trusted Platform Module. Hardware root-of-trust component used at Layer 0 of the provenance stack to provide hardware attestation for the actuator-layer signing key.

TriP. Trajectory Identity Protocol. The IETF Internet-Draft (draft-ayerbe-trip-protocol-03) defining the Proof-of-Trajectory mechanism for the GNS Protocol.

ULissy. The domain-specific programming language for moving machines that treats identity, location, time, cryptography and energy as first-class primitives. The propagates from construct of §12 is added in the present work.

VLA. Vision-Language-Action model. A class of multimodal model that takes visual and textual inputs and produces motor-command outputs for embodied systems.

Appendix B — References

Regulatory Instruments

- Regulation (EU) 2024/1689 of 13 June 2024 (Artificial Intelligence Act), OJ L of 12 July 2024.
- Regulation (EU) 2023/1230 of 14 June 2023 on machinery (Machinery Regulation), OJ L 165/1, 29 June 2023.
- Directive (EU) 2024/2853 of 23 October 2024 on liability for defective products, OJ L of 18 November 2024.
- Executive Order 14110 of 30 October 2023 (Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence), 88 FR 75191.
- 21 CFR Part 11 — Electronic Records; Electronic Signatures, US Food and Drug Administration.

Standards and Specifications

- Bernstein D. J., Duif N., Lange T., Schwabe P., Yang B.-Y. — Edwards-Curve Digital Signature Algorithm (EdDSA), RFC 8032, IETF, January 2017.
- Brodt R., Sambasivan S., et al. — H3: Uber’s Hexagonal Hierarchical Spatial Index, Uber Engineering, open-source release, 2018; current specification at h3geo.org.
- Coalition for Content Provenance and Authenticity (C2PA) — Technical Specification, version 2.0, 2024.
- Supply-chain Levels for Software Artifacts (SLSA) — Specification v1.0, OpenSSF, 2023.
- SPIFFE Specification, Cloud Native Computing Foundation, current version.
- World Wide Web Consortium — Verifiable Credentials Data Model 2.0, W3C Recommendation, 2024.
- Stellar Consensus Protocol, Stellar Development Foundation, current specification.

GNS Foundation Documents

- Ayerbe Posada C. — Trajectory Identity Protocol (TriP), IETF Internet-Draft draft-ayerbe-trip-protocol-03, 2026.
- GNS Foundation — GNS Protocol: Territorial Identity for AI Agents (GNS-AIP Case Study), v1.0, February 2026.
- ULISSY Foundation / GNS Protocol — ULissy: A Programming Language for Moving Machines, v0.1.0 Draft, 2025.
- USPTO Provisional Patent Application #63/948,788, GNS Foundation, 2025.

Industry and Research

- Google DeepMind — Gemini Robotics 1.5 and Embodied Reasoning 1.6: Technical Report, 2025.
- Intrinsic / Google — “Intrinsic joins Google to accelerate the future of physical AI,” 25 February 2026.
- Apptронik — Apollo deployment with Mercedes-Benz, public announcements, 2025–2026.
- Boston Dynamics — Atlas production roadmap, post-CES 2026 announcements.
- Agility Robotics / GXO — Digit deployment milestones, 2024–2026.
- Comau — Intrinsic Flowstate partnership announcement, 15 May 2023.

Cited Frameworks for Software-Agent Provenance

- Agent Action Reasoning Model (AARM) — academic publication, February 2026.
- Google — Agent Payments Protocol (AP2): Intent and Cart Mandates, September 2025.
- Prova — Attestation-as-a-service for AI agents, commercial product documentation, 2025–2026.

- Anjuna — TEE-based supervisor for agentic AI, commercial product documentation, 2025–2026.

Appendix C — Document Information

Document title. Propagation Identity: The Missing Primitive for Physical AI.

Document version. v0.1.0 — Draft, April 2026.

Authoring entity. GNS Foundation (Canton Zug, Switzerland) and ULISSY S.r.l. (Roma, Italy).

Companion documents. ULissy v0.1.0 Whitepaper (language); GNS-AIP v1.0 Case Study (AI agent identity); GNS Protocol v1.0 Whitepaper (human identity).

Standardization track. IETF, follow-on to draft-ayerbe-trip-protocol-03.

Patent posture. USPTO Provisional Patent #63/948,788 (parent invention). FRAND licensing for commercial implementations; non-assertion against open-source implementations.

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Comments and contributions. via the GNS Foundation contact channel; subsequent versions will incorporate community feedback ahead of v1.0.

“The journey is the proof.”

— GNS Protocol Design Principle

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